

Virtual Espionage: The Impact of Emerging Technology on Your Case

Thomas J. Daley, J.D.

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Introduction

Remember when espionage in a family dispute involved paying someone with a tolerance for boredom to sit outside a house and make notes, take photographs of people coming and going, maybe snap a few pictures through a window, and certainly rummaging through the trash for incriminating evidence?¹ While these methods still have value and are used by litigants in divorce and custody disputes² there are three important developments that broaden access to surveillance by replacing humans with machines, increase the invasiveness and evasiveness of spying techniques, and create self-imposed attack vectors exposed through the normal discovery process. This paper explores these developments and offers some ideas for reducing one's exposure to virtual espionage.

Note on Personal Safety

It is fun to consider these developments in virtual espionage in the context of spouses or co-parents behaving badly, sometimes comically badly, but there is a darker side to this. The same tools that will help a relatively innocent spouse detect the ill-conceived actions of the other can also be used by an abuser seeking to further

ensnare a victim or, worse, locate a victim for the purpose of imparting serious physical harm to that person. Kindly resist the rhetorical palliative of “I have nothing to hide so there’s nothing to worry about” and keep in mind that some of our clients are not hiding to avoid detection of poorly-considered acts—they are hiding for their lives.

Human vs. Machine

One limiting factor in surveillance is the cost.³ It is not easy to predict when someone will perform that one great act that blows the case wide open, such as leaving the children alone to go visit a paramour or inviting guests over for activities that are better off undisclosed. Considering the economic burden and the fact that much surveillance produces nothing but noise⁴, human-sourced spying is becoming a relic as machines become better at capturing and reporting what we want to know at a very low cost.

Threat: Geo-Trackers

Why pay to have someone followed when you can have the target follow themselves for less than \$50?

¹ Dan Nosowitz, *Private Eyes Tell Us About Digging Through People's Trash*, Atlas Obscura, Mar. 18, 2016, <https://www.atlasobscura.com/articles/private-eyes-tells-us-about-digging-through-peoples-trash>.

² Jack M. Riley, *Reconnaissance: Discovery – Ask and You Shall Receive* (State Bar of Texas Marriage

Dissolution 101, 2019), at 4, <https://www.texasbarcle.com/cle/OLViewArticle.aspx?a=205524&t=PDF&e=16897&p=1>.

³ *Id.*

⁴ *Id.*

Although there are other products elbowing for a piece of the market, Apple has attracted quite a bit of publicity over its AirTag device. AirTags are not trackers themselves. Instead they transmit a Bluetooth signal picked up by nearby iPhones that relay the location of the phone (and thus the AirTag) to Apple's network.⁵ Sounds great: place an AirTag on your keys, in your wallet, on your computer, in your child's pocket and never lose track again: Every time someone with an iPhone passes by the AirTag, Apple's private tag-wiki is updated with precise location information.

But all silver linings have a cloud and the specter on the horizon here is that anyone can slip an AirTag into anything they have access to and track it. For example, in New York, people have found AirTags in their bags, vehicles, and coat pockets.⁶ Meanwhile, law enforcement in Colorado, Georgia, Michigan, and Texas have been called upon to investigate AirTag misuse.⁷

AirTags and their ilk (GPS trackers, cellphones hidden in vehicles, Tile tags, etc.) pose a significant risk to people who do not wish to be located or tracked. Imagine a family violence victim who goes to the grocery store and while in the store the perpetrator inserts an AirTag in the vehicle. A person in Nashville, Tennessee, was curious about whether an AirTag was involved in the theft of his vehicle. When his vehicle was recovered, he asked local law enforcement to search the vehicle, which they

did and found nothing. The victim was notified by his mobile phone that someone was tracking him, so he decided to disassemble the car himself. In doing so, he found a slit in the weather-stripping around his door and an AirTag hidden away.⁸

Protection from Unauthorized AirTags

Apple has revised its iPhone-based app such that when it detects an AirTag that is not registered to that customer's iPhone, the phone presents a pop-up notification. Once the user has been alerted, the user can use Find My app to instruct the AirTag to play a sound so that it can be located.⁹

Apple publishes TrackerDetect for Android phone users. The TrackerDetect app will not provide automatic notifications for AirTags but can be used to scan for the presence of AirTags. Just like its iPhone-based relative, TrackerDetect can initiate a signal to the AirTag so that it plays a sound.¹⁰

Threat: Hidden Cameras

Cameras hidden where a surveillance subject does not expect them to be is a technique that many of us are familiar with. The more recent development in hidden cameras is their cost, miniaturization, and their ability to hide anywhere without a hardwired power source or network connection.

For example, a quick search on Amazon for "Best Sellers in Hidden Cameras"¹¹ brings up a \$36

⁵ *How Does it Work*, Apple, <https://www.apple.com/airtag/>.

⁶ Kashmir Hill, *I Used Apple AirTags, Tiles, and a GPS Tracker to Watch My Husband's Every Move*, New York Times, Feb. 11, 2022, <https://www.nytimes.com/2022/02/11/technology/airtags-gps-surveillance.html>.

⁷ David Ingram, *A Tracking Device Made by Apple is Showing Up in Suspected Crimes*, NBC News, Dec. 23, 2021, <https://www.nbcnews.com/news/apple-airtag-showing-up-crimes-rcna9416>.

⁸ Jeremy Finley, *Man Finds AirTag Police Couldn't*, WSMV4, Mar 24, 2022, <https://www.wsmv.com/2022/03/24/after-police-find-no-evident-airtag-theft-man-takes-apart-car-locate-it/>.

⁹ Jonathan Wylie, *Can AirTags Track You?*, Medium, Feb. 16, 2022, <https://jonathanwylie.medium.com/can-airtags-track-you-yes-4284f255804d>.

¹⁰ *Id.*

¹¹ See, <https://www.amazon.com/Best-Sellers-Hidden-Cameras/zgbs/photo/12909791>.

device that is a USB charging block and sports a tiny, status-LED-looking camera hidden, but not hidden, right above the USB port. They also showcase a \$39.98 product about the size of 5 quarters stacked together that gathers 1080p video and relays it via WiFi along with a similar but smaller product for \$19.99.

This technology is so cheap and easily available that one wonders whether one has a reasonable expectation to seclusion anywhere outside one's home or vehicle.¹²

Cameras such as these, hidden through one's home, can be used to gather evidence of bad acts or embarrassing photos to be used as "revenge porn."¹³

Detecting Hidden Cameras

These cameras are small and amenable to being tucked away in dark, powerless corners, and therefore can be difficult to detect. There are three detection techniques short of tearing the room down to the studs.

First, perform a physical search of the room. All cameras will have a glass lens and thus have at least a small reflective surface.¹⁴ One writer suggests darkening the room, and slowly scanning the room from multiple angles with a flashlight or a laser pointer and recommends that you search holes and gaps in walls and ceilings as well as HVAC vents.¹⁵

¹² See discussion of the *Vaughn* and *Kyllo* cases below.

¹³ For a thorough treatment of privacy between spouses, see, Heather L. King and Paul Leopold, *Spousal Privacy – Where It Begins and Where It Ends* (Advanced Trial Skills for Family Lawyers 2020) Dec 11, 2020.

¹⁴ Chelsea Ong, *Three Easy Ways to Find Hidden Cameras in Hotels and Rental Homes*, CNBC, Jun. 26, 2022, <https://www.cnbc.com/2022/06/27/how-to-find-a-hidden-spy-camera-in-hotel-rooms-and-rental-homes.html>.

¹⁵ Kim Komando, *How to Spot Hidden Surveillance Cameras in your Airbnb, VRBO, or Vacation Rentals*, USA TODAY, Jun. 23, 2022,

Next, look at the WiFi network using a WiFi scanning app such as "FING" to identify devices on the network. Cameras might be connected to a different network than the one published for your use, but the scanning apps will detect hidden and other WiFi networks. You should also look for devices that are broadcasting their own network name in search of a host app. Note that this technique will not work for cameras that record to an SD card or are otherwise not connected to a network.¹⁶

Finally, you can purchase a radio frequency (RF) detector. As with many products and services, at this point, one tends to get what one pays for: Inexpensive RF detectors deliver disappointing results while effective RF detectors can cost up to \$1,000.¹⁷

Threat: Cheap Cyberstalkers

Performing a search via Google for "tracking your spouse" brought up 18,300,00 results. The blogs exposed through that search recommend MySpy, an app that must be installed on a "jailbreak" Android device¹⁸ to take advantage of all the app has to offer or any iOS device.¹⁹

MySpy is marketed as a parental control app, but nothing, other than good sense, stops someone from using it for other purposes. The app allows the "parent" to monitor calls, track text messages, snoop in a large number of messaging apps such as WhatsApp, SnapChat, iMessage,

<https://www.usatoday.com/story/tech/columnist/komando/2022/06/23/how-check-hidden-cameras-airbnb-vrbo-vacation-rentals/7652726001/>.

¹⁶ Chelsea Ong, *Id.*

¹⁷ *Id.*

¹⁸ For an explanation of "jailbreaking," see, *How Does Jailbreaking or Rooting Affect My Mobile Device Security?*, McAfee, Jun. 13, 2012, <https://www.mcafee.com/blogs/mobile-security/how-does-jailbreaking-or-rooting-affect-my-mobile-device-security/>.

¹⁹ Product claim at <https://www.mspy.com/no-jailbreak.html>.

and more, track the phone's location, log keystrokes, record the phone's screen contents, view photos and videos on the phone, monitor all aspects of Internet browsing, read emails and calendars, and more.²⁰

Defense Against MySpy

The app claims to be undetectable and maximally invasive. If a person has any reason to believe that something such as MySpy is installed on their device, based on the product's claims, my suggestion would be to throw the phone away and get a new one.

Increase in Invasiveness and Evasiveness

It is well-established that people willingly expose more incriminating, or at least prejudicial, information about themselves on-line than would ordinarily be discoverable by a subpoena or court order.²¹ It has become commonplace for litigants and investigators to snoop through an opposing party's public social media posts, which are often rich sources of information.²² But with the rise of private profiles, secure platforms,

virtual private networks²³ and other obfuscation tools, how is one to know exactly what kind of socially nonnormative erotic proclivities one's counterparty is indulging in? For starters, try a keylogger.

Threat: Keyloggers

Traditionally, keyloggers are programs or apps that run on the surveillance subject's internet-enabled devices and capture the keys pressed on the keyboard.²⁴ It does not take much of an inferential leap to know that if you can record a person's every keystroke, you can replicate that series of keystrokes, and essentially replay on your computer screen what they were doing and seeing on their screen. Whether the subject's activities included prurient content on the most popular web site on the Internet²⁵, hiding financial resources, or diverting their romantic attention to a person not known to be part of the relationship in question, keylogging will reveal it all.

History of Keylogging

Keylogging has been around since at least the mid-1970s when the Soviet Union developed a

²⁰ <https://www.mspy.com/features.html>.

²¹ See, e.g., *Largent v. Reed*, No 2009-1823, 2011 WL 5632688 (Pa.Com.Pl. Nov. 08, 2011) (Holding that litigant's Facebook posts, which controverted her personal injury claims, were probative, relevant, and discoverable), available at <https://www.theemployerhandbook.com/files/2015/01/Largent.pdf>.

²² See, e.g., Rachel Levinson-Waldham, et al, *Social Media Surveillance by the U.S. Government*, Brennan Center for Justice, Jan 7, 2022, <https://www.brennancenter.org/our-work/research-reports/social-media-surveillance-us-government> ("Social media has become a significant source of information for U.S. law enforcement and intelligence agencies. The Department of Homeland Security, the FBI, and the State Department are among the many federal agencies that routinely monitor social platforms, for purposes ranging from conducting investigations to identifying threats to screening travelers and immigrants.").

²³ For a full explanation of the uses and benefits of virtual private networks, see, National Institute of Standards and Technology, Special Publication 800-113, *Guide to SSL VPNs: Recommendations of the National Institute of Standards and Technology* (2008), available at <https://nvlpubs.nist.gov/nistpubs/Legacy/SP/nistspecialpublication800-113.pdf>.

²⁴ Nikolay Grebennikov, *Keyloggers: How They Work and How to Detect Them (Part 1)*, Securelist, Mar 29, 2007, <https://securelist.com/keyloggers-how-they-work-and-how-to-detect-them-part-1/36138/>.

²⁵ Lisa Baker, *Pornhub Receives More Website Traffic Than Amazon and Netflix, New Research Reveals!*, Business In the News, Jul. 19, 2020, citing Fasthosts.co.uk, <https://businessinthenews.co.uk/2020/07/19/pornhub-receives-more-website-traffic-than-amazon-and-netflix-new-research-reveals/>.

digital bugging device that could be hidden in IBM Selectric typewriters.²⁶ This first known method for capturing keystrokes involved modifications to the typing device itself in a manner analogous to keylogger apps and programs that run on the computer being surveilled today. With over fifty years of device-modification-based keylogging history, numerous defenses have been developed.

Defense Against Traditional Keyloggers

The simplest defense against traditional keylogger programs and apps is commercial malware software. Once loaded, malware protection software will periodically scan the host device for the digital fingerprints of thousands of digital bad actors, including keyloggers.²⁷

Another defense is to run a reverse-firewall²⁸ that pops up an alert anytime an unknown program attempts to make a network connection. This does not stop keyloggers, per se, but it is an impediment to the clandestine app's ability to relay the captured keystrokes to a remote computer, such as that of a traditional PI or a motivated spouse.

A nearly²⁹ fool-proof defense is to reboot the target computer from removable media that is known to be clean and safe. This ensures that any malicious or unwanted apps installed on the target computer are absent. Of course, this method of protection requires one to deviate somewhat from the conduct of the other cool kids by carrying around a USB dongle with a bootable version of Windows, iOS, or Linux installed.

Recent Developments in Keyloggers

One characteristic of all keyloggers discussed up to this point is that the person seeking to capture keystrokes needs to gain access to the subject device. Whether that is an Eastern Bloc spy modifying a typewriter from fifty years ago to a suspicious spouse who recently downloaded a keylogging app, at some point the spy needs access to the device, whether that is physical access or remote access. But what about logging keystrokes when one does not have any access to the subject device?

For at least the last seven years, researchers have applied novel uses of radio signals to capture keystrokes.³⁰ The algorithms are mindbogglingly complex³¹, particularly to practitioners of a profession who frequently brag

²⁶ Sharon A Maneki, *Learning from the Enemy: The GUNMAN Project*, Center for Cryptologic History, National Security Agency, 2018 (2d ed.), <https://media.defense.gov/2021/Jul/13/2002761779/-1/-1/0/LEARNINGFROMTHEENEMYGUNMAN.PDF>.

²⁷ Twinkle Arora, et al, *Detection of Cyber Crime on Social Media using Random Forest Algorithm*, (2019 2nd International Conference on Power Energy, Environment and Intelligent Control (PEEIC)), Oct. 19, 2019, at 47-51, <https://ieeexplore.ieee.org/document/8976474>.

²⁸ E.g., *Windows Defender Firewall with Advanced Security*, Microsoft, Jun. 2, 2022, <https://docs.microsoft.com/en-us/windows/security/threat-protection/windows-firewall/windows-firewall-with-advanced-security>.

²⁹ A determined spy could install a keylogger at the BIOS level, in which case none of these defenses

would be effective. If one finds that one is in a relationship with someone who is capable of installing a BIOS-based keylogger, it is time to grab the "go bag" and walk away slowly—one has found an enemy who cannot be beat.

³⁰ Kamran Ali, et al, *Keystroke Recognition Using WiFi Signals* (MobiCom '15: Proceedings of the 21st Annual International Conference on Mobile Computing and Networking) Sep. 7, 2015, at 90-102, <https://dl.acm.org/doi/10.1145/2789168.2790109>.

³¹ See, Yi Zheng, et al, *Through-Wall Tracking with Radio Tomography Networks Using Foreground Detection* (2012 IEEE Wireless Communications and Networking Conference (WCNC)) 2012, at 3280, <https://span.ece.utah.edu/uploads/zheng-wcnc2012-RT-nets-foreground-detection.pdf>.

that they went to law school to avoid having to apply math, but some simple analogies can clarify the techniques.

Imagine a person standing behind a white curtain, with backlighting. The background light would shine through except where the person was standing—we would call that area of occlusion a “shadow.” When the person moved their right arm, we would see a distinct change in the shape of the shadow. Likewise, when the person moved their left arm, we would see a different and distinct change in the shape of the shadow. With this technique, from a great distance, so long as we, or a video camera aimed in that direction, maintained line of sight with the backlit curtain, we would know with a high degree of certainty what the person was doing.

Now imagine that instead of light and shadows being detected and interpreted by a human eyeball and brain the source of illumination was ambient WiFi signals leaking out from a building or cellular radio signals permeating a building. These ambient radio waves are not detectable by the human eye, but we know they are detectable by computers because computers ride those waves to send and receive data over the Internet. From experience we know that objects in a building influence the strength of radio waves passing through that building—there are areas of every building where the WiFi signal is weak or cell phones struggle to sustain a quality connection.

When people move about buildings, they cast “shadows” in the radio signals just like they do when they pass between an observer and a source of visible light. And engineers from around the world have developed algorithms to

read those shadows in ambient radio waves and detect minute motions, including typing on keyboards. Does this mean a person sitting in your parking lot or driveway can read your keystrokes without ever gaining access to the inside of the building or to the subject device? YES!

Why the detailed explanation in a family law CLE article? Because this is where electronic surveillance is headed and pathologically curious adversaries are going to have even more information than that which is provided through the social media firehose they drink from currently. In 2015 the technology was about 94% accurate and ran on an off-the-shelf laptop hosting an app called WiKey.³²

Defenses Against Radio-Signal-Based Keyloggers

The signals being measured are weak and the test conditions are well-controlled.³³ Therefore, one defense would be to work in a room with other random movement, e.g. in a coffee shop with many other people moving around.³⁴ Another defense would be to move about the building so that not all clandestine computer work is done in the same room with the same radio spectrum profile. Finally, the current machine-learning-based tools require training for each individual typist and environment.³⁵ Knowing that, the subject typist can avoid providing a useful baseline by making sure app-based keyloggers are not installed and by moving from room to room when engaging in digital conduct.

³² Kamran Ali, et al, at 90, (“In real-world experiments, WiKey can recognize keystrokes in a continuously typed sentence with an accuracy of 93.5%”).

³³ *Id.* at 100-101.

³⁴ Although working in public exposes the surreptitious typist to an even older (i.e. more

mature) acoustic-based remote keylogger. See, Dmitri Asonov, et al, *Keyboard Acoustic Emanations* (2012 IEEE Symposium on Security and Privacy) 2004, at 3.

³⁵ Kamran Ali, et al, at 101.

Threat: Radio-Based People Locators

The U.S. Department of Justice has envisioned through-wall surveillance since as early as 2006.³⁶ The technology tested by DOJ relied on sonar, when looking behind metallic walls, and a combination of sonar and radar when looking through non-metallic walls.³⁷ The deployment of radar and sonar may have been a limitation on the contractor's imagination in 2006. It could also have been a response to the *Kyllo* case discussed below in that the technology would not rely on the passive interception of information produced inside someone's residence but would instead rely on signals originating from the surveillance team and, theoretically, deployed pursuant to a warrant.

Then the universities got involved. It started as an innocent idea. Remember the discussion above about radio shadows? In 2013 MIT developed software using similar techniques to track people inside buildings.³⁸ The idealized applications of this related to gaming and fall detection among the elderly³⁹ and eventually led to the introduction of a product called Walabot. Walabot could detect motion through obstacles and call for assistance if a fall or lack of motion was detected by the product.⁴⁰

Walabot "tracks the location, movement, posture, and vital signs of people in the home—and is so sensitive that it can even monitor breathing during sleep."⁴¹

Imagine Walabot, or a successor product, being left behind by a jilted spouse or an abuser just before being kicked out through a protective order. The innocent-sounding features could be weaponized to count the number of people in the house, locate them within the space, and inform the remote person when someone was at their most vulnerable – alone and asleep.

In another twist, an Israeli startup is offering a device to the military that allows soldiers to see through walls in three dimensions using radar.⁴²

Hidden cameras, keyloggers, spyware, and private investigators are familiar elements of highly contested relationships. But ask people under the protection of a protective order or hiding at a shelter how safe they would feel knowing that all the walls surrounding them were made of transparent glass and the question would likely seem ridiculous.

Yet, the technology is here, the products are affordable, and they are becoming more capable with every iteration. There may be little reason to worry about through-wall person tracking being ubiquitous or even present in one's cases today, but it will be and we are not likely to know when until we are confronted by it.

Is Radio-Signal Tracking Legal?

One deterrent to people employing these developing, invasive technologies would involve

³⁶ U.S. Department of Justice, Report Number 13620-FR-01, *Through-Wall Surveillance for Locating Individuals Within Buildings* (2006).

³⁷ *Id.* at 13.

³⁸ Abby Abazorius, *New System Allows for High-Accuracy, Through-Wall, 3-d Motion Tracking*, MIT News, Dec. 11, 2013, <https://news.mit.edu/2013/new-system-allows-for-high-accuracy-through-wall-3-d-motion-tracking-1211>.

³⁹ *Id.*

⁴⁰ Matthew Greenwood, *Smart Home Sensor Sees Through Walls*, Engineering.Com, Oct. 25, 2018, <https://www.engineering.com/story/smart-home-sensor-sees-through-walls>.

⁴¹ *Id.*

⁴² Bharat Sharma, *Bizarre New Military Tech Lets Soldiers See Through Walls, Objects in Any Setting*, India Times, Jun 28, 2022, <https://www.indiatimes.com/technology/science-and-future/bizarre-new-military-tech-lets-soldiers-see-through-walls-objects-in-any-setting-573342.html>.

the threat of civil damages or criminal prosecution.

The technology is so new that specific case law could not be located. However, Texas recognizes the tort of “intrusion of seclusion.”⁴³

In the *Vaughn* case, the Court acknowledged a general right to privacy in Texas. The Court further taught that determining if an invasion of seclusion had occurred turns on whether the alleged tortfeasor invaded a private domain wherein which the victim could claim a reasonable expectation of seclusion. In *Vaughn*, the plaintiff complained that the defendant looked at her using binoculars when she was standing outside and when she was standing in her kitchen in front of an open window. Notwithstanding its acknowledgment of a right to seclusion, the Court found no tortious invasion of seclusion because it was unreasonable for a person to expect seclusion to include standing outside or in front of an open window.⁴⁴ Thus, waves within the electromagnetic spectrum that are visible to the unaided human eye, i.e. visible light, and, presumably, mechanical analogs such as cameras, are not protected from detection outside the residence.⁴⁵

However, in the *Kyllo* case, the Supreme Court of the United States considered whether the detection of electromagnetic waves outside the visible spectrum through the use of a thermal imager directed toward a private home

constituted a search under the Fourth Amendment. In *Kyllo*, a park ranger suspected that Danny Kyllo was growing marijuana in his home. (Apparently growing marijuana indoors requires high-intensity lamps for an optimal growing environment.) The ranger had the novel idea of using a thermal imager for a few seconds from across the street to see whether Kyllo’s house emitted an abnormally high heat signature. Based on a determination that Kyllo’s house was emitting quite a bit of extra heat, combined with information from an informant, the ranger obtained a search warrant, searched the home, and sure enough found a marijuana farm inside. Justice Scalia, writing for the Court, reasoned that “obtaining by sense-enhancing technology any information regarding the interior of the home that could not otherwise have been obtained without physical ‘intrusion into a constitutionally protected area,’ constitutes a search—at least as where (as here) the technology in question is not in general public use.”⁴⁶

The *Kyllo* opinion makes it clear that when emanating from a location wherein which a person could claim a reasonable expectation of seclusion, electromagnetic waves that cannot be detected without “sense-enhancing tech” are constitutionally protected under the Fourth Amendment and, by application of the reasoning in *Vaughn*, collecting and analyzing those waves would constitute a tortious invasion of seclusion in Texas.

⁴³ Kevin T. Segler, *Use of “New Age” Discovery Tools in Family Law Cases* (Family Law 101) Jul. 31, 2016, at 6; *Billings v. Atkinson*, 489 S.W.2d 858, 860 (Tex. 1973); *Texas State Employees Union v. Texas Dep’t of Mental Health and Retardation*, 746 S.W.2d 203, 205 (Tex. 1987).

⁴⁴ *Vaughn v. Drennon*, 202 S.W.3d 308, 320 (Tex. App.—Tyler 2006, no pet.) (“The core of the tort of invasion of privacy is the offense of prying into the private domain of another. [...] One cannot expect to be entitled to seclusion when standing directly in

front of a large window with the blinds open or while outside.”).

⁴⁵ *Rf, e.g., Arnold Pompos, Why Can Radio Waves Pass Through a Wall but Light Cannot*, Fermi Research Alliance, LLC, Apr. 28, 2014, <https://www.fnal.gov/pub/science/inquiring/questions/mikep.html>.

⁴⁶ *Kyllo v. United States*, 533 US 27 (2001).

Rf, eg, Arnold Pompos, Why can Radio Waves Pass Through a Wall But Light Cannot, Fermi Research Alliance, LLC, Apr. 28, 2014

Self-Imposed Attack Vectors

Many of us have read the term, “digital footprints.” Used in the context of our daily lives, digital footprints are the persistent electronic memorializations of activities that are enabled, enhanced, and recorded by digital devices. Our digital footprints leave traces of our activities and former whereabouts with the same clarity as warm boots marching through cold snow—but the snow never melts and the footprints never disappear.

So far, this paper has focused on surveillance that can be done without our knowledge or willing participation. But sometimes we knowingly take a hike through that snow and are disappointed—perhaps mortified—when someone sees our tracks seconds, days, or years later. Here are some common activities that create significant sources of information and that are purely optional and under our control.

Threat: Toll Tags

We all know that cashless tolling authorities, such as NTTA, collect and retain historical trip data, including “usage, or driving times and patterns and GPS location(s).”⁴⁷ The data gathered by these agencies in Texas is presumed by the agencies to be public information, subject

to some limitations.⁴⁸ Knowing that one’s use of a cashless toll app is tracked and available should be baked into everyone’s plans: Do not use toll roads when you are on a secret mission.

What is probably not factored into many people’s behavior is the tracking of one’s toll tag when not using a toll road. In 2015, a hacker in New York modified his E-Z Pass to flash a light and make a sound whenever the E-Z Pass device was being read.⁴⁹ This usage, combined with NTTA’s privacy policy suggests that NTTA could be, and certainly is capable of, tracking and storing vehicle usage on surface streets.

Defenses Against Toll Tag Tracking

The most obvious defense is to not use tollways and to not travel with a cashless toll collection transponder. If that is not practical, there is at least one product on the market that purports to alert drivers to hidden toll transponders, assuming the transponders use the same frequency and signaling protocol of E-ZPass.^{50, 51}

Threat: Cell Phone Location History

Your cell phone tracks your location whenever it is with you and powered on. Everyone knows that. What is not common knowledge is that cell phone carriers sell your ping (location) data to data analytics companies.⁵²

⁴⁷ North Texas Tollway Authority, *Privacy Policy*, Jan 1, 2022, <https://www.ntta.org/Pages/Privacy-Policy.aspx>.

⁴⁸ North Texas Tollway Authority, *Public Information Act*, (no date), <https://www.ntta.org/newsresources/publicinfoact/Pages/default.aspx>.

⁴⁹ Scott Schober, *Is Your E-Zpass Spying On You?*, Medium, Apr. 9, 2016, <https://medium.com/@ScottBVS/is-your-e-zpass-spying-on-you-53e32ed25808>.

⁵⁰ See Detect-A-Pass E-Z Pass Detector sold for \$99 at <https://scottsschober.com/product/detect-a-pass-e-z-pass-detector/>.

⁵¹ Note that toll collection points often employ license plate readers that collect discoverable data. See,

Carrie Rogers, *NTTA Customer Experience: The Next Level*, North Texas Tollway Authority, Oct. 8, 2013, <https://www.ntta.org/whatwedo/govtaff/Documents/Custom%20Service%20Enhancements%20-%20Detail%20Presentation.pdf>. If you doubt the (potential) ubiquity of license plate readers, see, e.g., *Car Plate Recognition System with Raspberry Pi and Node-RED*, Random Nerd Tutorials, Apr. 2, 2019, <https://randomnerdtutorials.com/car-plate-recognition-system-with-raspberry-pi-and-node-red/>.

⁵² Jennifer Valentino-DeVries, *How Your Phone is to Track You, and What You Can Do About It*, New York Times, Aug 19, 2020, <https://www.nytimes.com/2020/08/19/technology/smartphone-location-tracking-opt-out.html>; Federal

Your cell phone account, whether through Google or Apple, permits you to download your location history in detail.⁵³ When you are in a legal dispute and your whereabouts are probative of a disputed issue, this ability to download data is a blessing and a curse because if you can download it, you have possession of it.⁵⁴ If your location history is relevant⁵⁵ then your opposing party is entitled to a copy of your location history and you will have to produce it upon request.⁵⁶

Defenses to Embarrassing Location Histories

One having travel habits that could call one's judgment, veracity, fidelity, or ability to parent into question, may not want to maintain a detailed log of their whereabouts. Three ways to do that are to (1) power off all your devices (phones, fitness trackers, etc.) during secret trips; (2) place all those items in a Faraday bag, which will block all signals into and out of the enclosed devices⁵⁷; or (3) leave the devices—all of them—at home.

Threat: Good Ol' Social Media

Social media usage is the virtual surveillance fountain that never runs dry. We advise clients

to absolutely, 100%, full stop quit using social media while their case is pending.

Defenses to Social Media Mishaps

Here are some tips from well-known Family Law attorneys in Texas.⁵⁸

1. Archive your entire social media account, one by one, and provide those archives to your attorney for safekeeping to avoid a spoliation claim.
2. Consider taking down your social media pages—after archiving. If that is unbearable, weed out “friends” who you do not know well in real life.
3. Adjust your profile settings to the highest possible privacy settings.
4. Remove “friends” you do not actually know and do not accept friend requests from anyone you do not know well in real life.
5. Make yourself invisible by selecting “only friends” under search visibility.
6. After archiving, take down all your photos and untag yourself. You should not have any more than a simple, dignified headshot.

Communications Commission, FCC PROPOSES OVER \$200 MILLION IN FINES AGAINST FOUR LARGEST WIRELESS CARRIERS FOR APPARENTLY FAILING TO ADEQUATELY PROTECT CONSUMER LOCATION DATA, Feb. 28, 2020, <https://docs.fcc.gov/public/attachments/DOC-362754A1.pdf>.

⁵³ Sandy Writtenhouse, *How to Download Your Google Maps Data*, HowToGeek, May 11, 2021, <https://www.howtogeek.com/725241/how-to-download-your-google-maps-data/>.

⁵⁴ TEX. R. CIV. P. 192.7(b) (“Possession, custody, or control of an item means that the person either has physical possession of the item or has a right to possession of the item that is equal or superior to the person who has physical possession of the item.”).

⁵⁵ TEX. R. EVID. 401 (“Evidence is relevant if: (a) it has any tendency to make a fact more or less probably

than it would be without the evidence; and (b) the fact is of consequence in determining the action.”).

⁵⁶ TEX. R. CIV. P. 192.3(b) (“A party may obtain discovery of the existence, description, nature, custody, condition, location, and contents of documents and tangible things (including papers, books, accounts, drawings, graphs, charts, photographs, electronic or videotape recordings, data, and data compilations) that constitute or contain matters relevant to the subject matter of the action. A person is required to produce a document or tangible thing that is within the person's possession, custody, or control.”)

⁵⁷ Matt Jancer, *Get Lost With a Signal-Blocking Smartphone Pouch*, Wired, Sep. 22, 2020, [wired.com/story/silent-pocket-signal-blocking-smartphone-pouch/](https://www.wired.com/story/silent-pocket-signal-blocking-smartphone-pouch/).

⁵⁸ Charlie Jones, et al, *SOCIAL MEDIA – Dos and Don'ts for Our Clients*, TAFLS Trial Institute, Feb 3, 2022.

7. Assume that everything you submit to a social media app or web site will be viewed by the other attorney and the judge.
8. Understand that your counterparty in the case is very likely to require you to submit your entire social media profiles to them in discovery.
9. Do not assume you are safe. Even the best friends can be sources of leaks to others who are working with the other side of your case.
10. Do not send any email, DM, posts, or messages about your case.
11. Do not join chat groups. Your data are not private, they are owned by the network, and the data are highly searchable.
12. Do not forget that what goes on the Internet, stays on the Internet forever.
13. Do not make any posts or send any messages in anger.
14. Do not hit "Reply-All" until you triple check who "all" is.
15. Do not post a picture you would not want to show the judge in open court.

Threat: Fitness Trackers, Doorbell Cameras

John Browning recently wrote about the discovery perils of doorbell cameras and fitness trackers.⁵⁹ For example, if a spouse is claiming a need for post-divorce maintenance or a disproportionate share of the marital estate due to physical disability, that spouse's fitness tracker data would meet the test for relevance under TRE 401, the data would certainly be in the claiming spouse's possession as defined by TRCP 192.7(b), and thus the data would be discoverable under TRCP 196.1.

Likewise, imagine a party claiming that the other party is frequently late in arriving to the door for

exchanges of the children or that the party-in-residence forces the traveling party to wait an inordinate amount of time during exchanges. If there is a dispute about arrival times or conduct during exchanges, then the door camera recordings are relevant under TRE 401, they are in the possession of the party in control of the residence under TRCP 192.7, and thus they are discoverable from the party in control of the account containing the videos under TRCP 196.1.

Think about that. Every data-hoarding device owned by a party, from fitness trackers, to phones, to security cameras, to social media history, to browser history is discoverable if it meets the test for relevance.

Threat: Not Knowing How to Use Your Technology

I have added this threat mostly for the novelty of it, as it relates to my experience, and what some might see as the humor of the situation.

I was recently involved in a divorce case that arose from one party discovering the other party's infidelity. That's not the novel part. What sets this case apart is this: The spouses were on vacation with family and spent the day touring around and taking photographs. When they returned to the hotel that evening, everyone stood around, iPhones in hand, flinging photos to each other via AirDrop. Suddenly, the unfaithful spouse accidentally dropped to the innocent spouse and others in the family group a video vividly documenting a recent act of infidelity in full color with surround sound.

The day was ruined.

The unfaithful spouse was sent packing and the innocent spouse was on a Zoom consultation with a divorce attorney the next day.

⁵⁹ John G. Browning, "Fit" to Admit? Emerging Sources of Digital Evidence, *The Advocate*, vol. 96, p. 16.

Defenses Against Self-Harm

1. Do not commit bad acts.
2. If you do commit a bad act, do not make a video of it.
3. If you make a video of it, at least be sufficiently familiar with your technology that you can avoid sharing the video with your spouse and extended family.
4. If you do not take the time to learn your technology, then make flexible travel plans in case a vacation ends ignominiously. Sometimes an inadvertent flick of a finger across the glass plane of an ever-present cell phone is all it takes to make the quantum transition from happily married to mired in divorce litigation.